```
clc;
clear;
close all;

%% ============================================================
% CHARACTERISTICS STUDY OF SIDEBAND POWER AND FREQUENCIES IN AM
% USING MATLAB
% ============================================================

%% 1. INITIAL PARAMETERS

Fs = 1e6;              % Sampling frequency = 1 MHz
T  = 0.01;             % Simulation time = 10 ms

t = 0:1/Fs:T-1/Fs;

% Carrier and message parameters
fc = 100e3;            % Carrier frequency = 100 kHz
fm = 10e3;             % Message frequency = 10 kHz

Ac = 1;                % Carrier amplitude
Am = 1;                % Message amplitude

R = 50;                % Load resistance = 50 ohms

%% 2. MODULATION INDEX

mu = Am/Ac;

fprintf('\n');
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('       AM SIDEBAND POWER AND FREQUENCY STUDY\n');
```

```
       AM SIDEBAND POWER AND FREQUENCY STUDY
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('\nBasic Parameters:\n');
```

```
Basic Parameters:
```

```
fprintf('Carrier Frequency       = %.2f kHz\n',fc/1000);
```

```
Carrier Frequency       = 100.00 kHz
```

```
fprintf('Message Frequency       = %.2f kHz\n',fm/1000);
```

```
Message Frequency       = 10.00 kHz
```

```
fprintf('Carrier Amplitude       = %.2f V\n',Ac);
```

```
Carrier Amplitude       = 1.00 V
```

```
fprintf('Message Amplitude       = %.2f V\n',Am);
```

```
Message Amplitude       = 1.00 V
```

```
fprintf('Load Resistance         = %.2f Ohm\n',R);
```

```
Load Resistance         = 50.00 Ohm
```

```
fprintf('Modulation Index        = %.2f\n',mu);
```

```
Modulation Index        = 1.00
```

```

%% 3. MESSAGE SIGNAL

message = Am*cos(2*pi*fm*t);

%% 4. CARRIER SIGNAL

carrier = Ac*cos(2*pi*fc*t);

%% 5. AM SIGNAL

AM_signal = Ac*(1 + mu*cos(2*pi*fm*t)) ...
            .* cos(2*pi*fc*t);

%% ============================================================
% 6. TIME DOMAIN PLOTS
% ============================================================

% Message signal

figure;

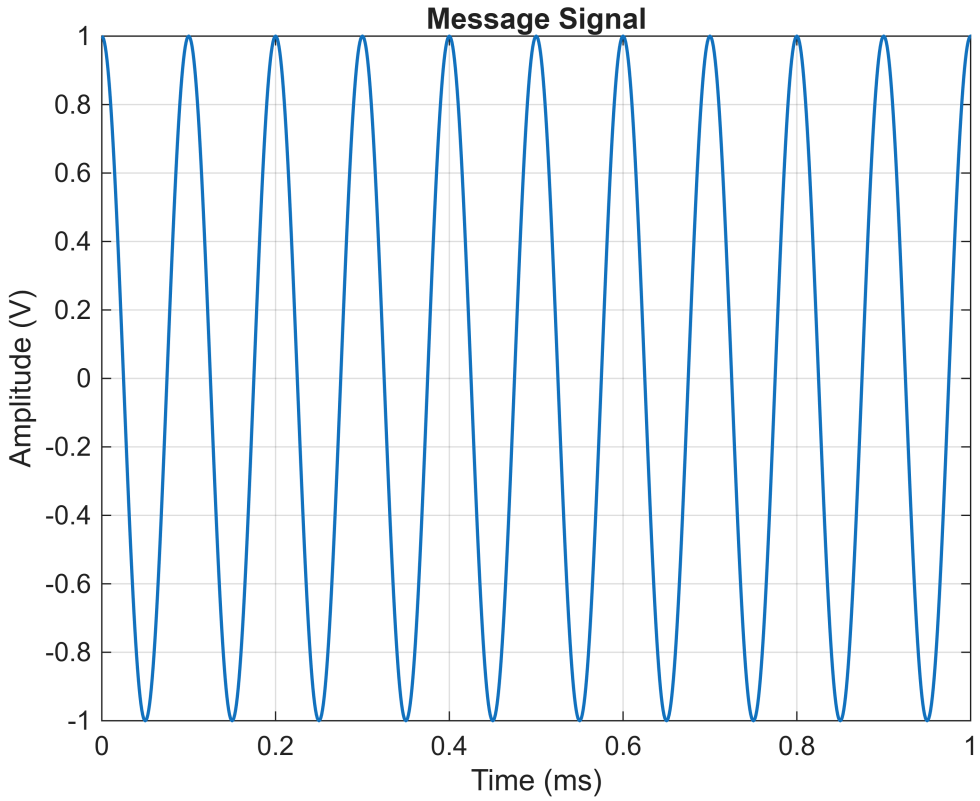
plot(t*1000,message,'LineWidth',1.2);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Message Signal');

grid on;

xlim([0 1]);
```




```
% Carrier signal

figure;

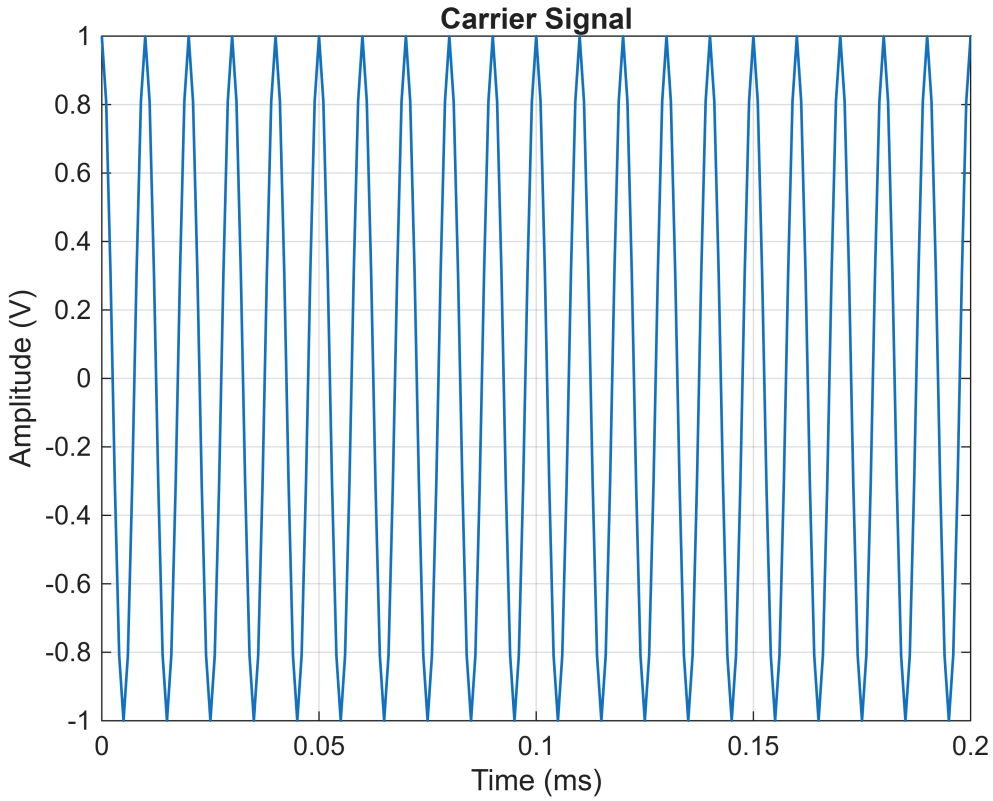
plot(t*1000,carrier,'LineWidth',1);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Carrier Signal');

grid on;

xlim([0 0.2]);
```




```
% AM signal

figure;

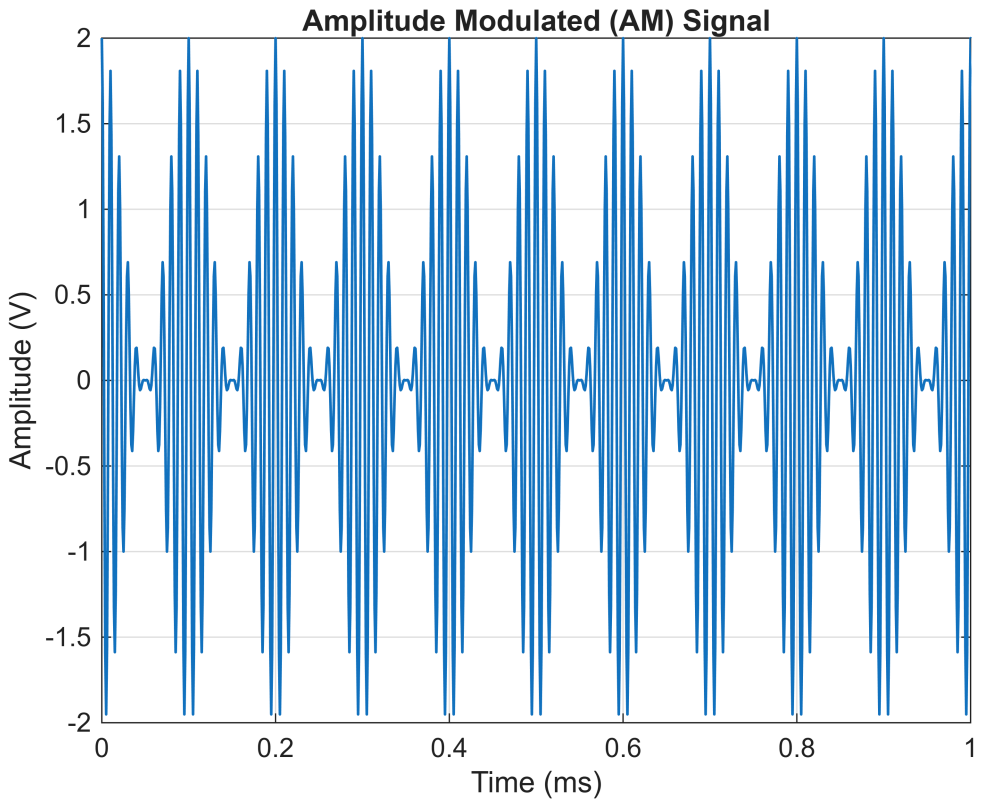
plot(t*1000,AM_signal,'LineWidth',1);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Amplitude Modulated (AM) Signal');

grid on;

xlim([0 1]);
```




```
%% ============================================================
% 7. AM SPECTRUM
% ============================================================

N = length(AM_signal);

AM_fft = fft(AM_signal);

AM_fft_shifted = fftshift(AM_fft);

f = (-N/2:N/2-1)*(Fs/N);

AM_magnitude = abs(AM_fft_shifted)/N;

figure;

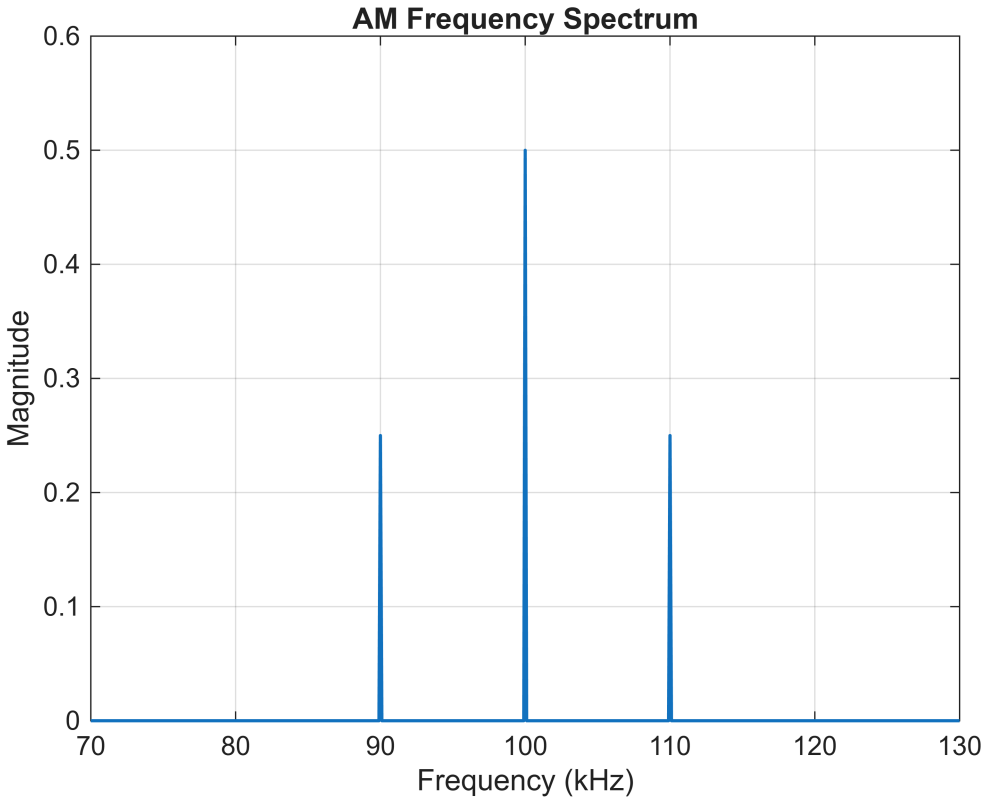
plot(f/1000,AM_magnitude,'LineWidth',1.2);

xlabel('Frequency (kHz)');
ylabel('Magnitude');

title('AM Frequency Spectrum');

grid on;
```

```
xlim([70 130]);
```




```
%% ============================================================
% 8. FIND USB AND LSB FREQUENCIES
% ============================================================

f_USB = fc + fm;

f_LSB = fc - fm;

fprintf('\n');
fprintf('Sideband Frequencies:\n');
```

```
Sideband Frequencies:
```

```
fprintf('LSB Frequency            = %.2f kHz\n',f_LSB/1000);
```

```
LSB Frequency            = 90.00 kHz
```

```
fprintf('Carrier Frequency        = %.2f kHz\n',fc/1000);
```

```
Carrier Frequency        = 100.00 kHz
```

```
fprintf('USB Frequency            = %.2f kHz\n',f_USB/1000);
```

```
USB Frequency            = 110.00 kHz
```

```
%% ============================================================
% 9. BANDWIDTH
% ============================================================

BW_AM = 2*fm;

fprintf('\n');
fprintf('AM Bandwidth:\n');
```

```
AM Bandwidth:
```

```
fprintf('BW = 2fm = %.2f kHz\n',BW_AM/1000);
```

```
BW = 2fm = 20.00 kHz
```

```
%% ============================================================
% 10. CARRIER POWER
% ============================================================

Pc = Ac^2/(2*R);

fprintf('\n');
fprintf('Carrier Power:\n');
```

```
Carrier Power:
```

```
fprintf('Pc = %.6f W\n',Pc);
```

```
Pc = 0.010000 W
```

```
%% ============================================================
% 11. SIDEband POWER
% ============================================================

P_USB = (mu^2 * Pc)/4;

P_LSB = (mu^2 * Pc)/4;

P_SB_total = P_USB + P_LSB;

%% ============================================================
% 12. TOTAL AM POWER
% ============================================================

P_total = Pc + P_SB_total;
```

```
%% ============================================================
% 13. POWER EFFICIENCY
% ============================================================

efficiency = (P_SB_total/P_total)*100;

%% ============================================================
% 14. DISPLAY POWER RESULTS
% ============================================================

fprintf('\n');
fprintf('Power Calculations:\n');
```

```
Power Calculations:
```

```
fprintf('Carrier Power             = %.6f W\n',Pc);
```

```
Carrier Power           = 0.010000 W
```

```
fprintf('USB Power                 = %.6f W\n',P_USB);
```

```
USB Power               = 0.002500 W
```

```
fprintf('LSB Power                 = %.6f W\n',P_LSB);
```

```
LSB Power               = 0.002500 W
```

```
fprintf('Total Sideband Power      = %.6f W\n',P_SB_total);
```

```
Total Sideband Power     = 0.005000 W
```

```
fprintf('Total AM Power            = %.6f W\n',P_total);
```

```
Total AM Power          = 0.015000 W
```

```
fprintf('Transmission Efficiency   = %.2f %%\n',efficiency);
```

```
Transmission Efficiency  = 33.33 %
```

```
%% ============================================================
% 15. MODULATION INDEX STUDY
% ============================================================

mu_values = 0.1:0.1:1.0;

P_USB_values = zeros(size(mu_values));

P_LSB_values = zeros(size(mu_values));
```

```
P_SB_values = zeros(size(mu_values));

P_total_values = zeros(size(mu_values));

Efficiency_values = zeros(size(mu_values));

for k = 1:length(mu_values)

    m_index = mu_values(k);

    P_USB_values(k) = ...
        (m_index^2 * Pc)/4;

    P_LSB_values(k) = ...
        (m_index^2 * Pc)/4;

    P_SB_values(k) = ...
        P_USB_values(k) + P_LSB_values(k);

    P_total_values(k) = ...
        Pc + P_SB_values(k);

    Efficiency_values(k) = ...
        (P_SB_values(k)/P_total_values(k))*100;

end

%% ============================================================
% 16. DISPLAY MODULATION INDEX TABLE
% ============================================================

fprintf('\n');
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('      POWER VS MODULATION INDEX\n');
```

```
      POWER VS MODULATION INDEX
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```

fprintf('\n');

fprintf(' Mu     USB Power    LSB Power    Total SB     Total AM\n');
```

```
  Mu     USB Power    LSB Power    Total SB     Total AM
```

```
fprintf('-----------------------------------------------------\n');
```

```
-------------------------------------------------
```

```
for k = 1:length(mu_values)

    fprintf(' %.1f     %.6f     %.6f     %.6f     %.6f\n', ...
        mu_values(k), ...
        P_USB_values(k), ...
        P_LSB_values(k), ...
        P_SB_values(k), ...
        P_total_values(k));

end
```

```
 0.1     0.000025     0.000025     0.000050     0.010050
 0.2     0.000100     0.000100     0.000200     0.010200
 0.3     0.000225     0.000225     0.000450     0.010450
 0.4     0.000400     0.000400     0.000800     0.010800
 0.5     0.000625     0.000625     0.001250     0.011250
 0.6     0.000900     0.000900     0.001800     0.011800
 0.7     0.001225     0.001225     0.002450     0.012450
 0.8     0.001600     0.001600     0.003200     0.013200
 0.9     0.002025     0.002025     0.004050     0.014050
 1.0     0.002500     0.002500     0.005000     0.015000
```

```
fprintf('-----------------------------------------------------\n');
```

```
-------------------------------------------------
```

```
%% ============================================================
% 17. USB POWER VS MODULATION INDEX
% ============================================================

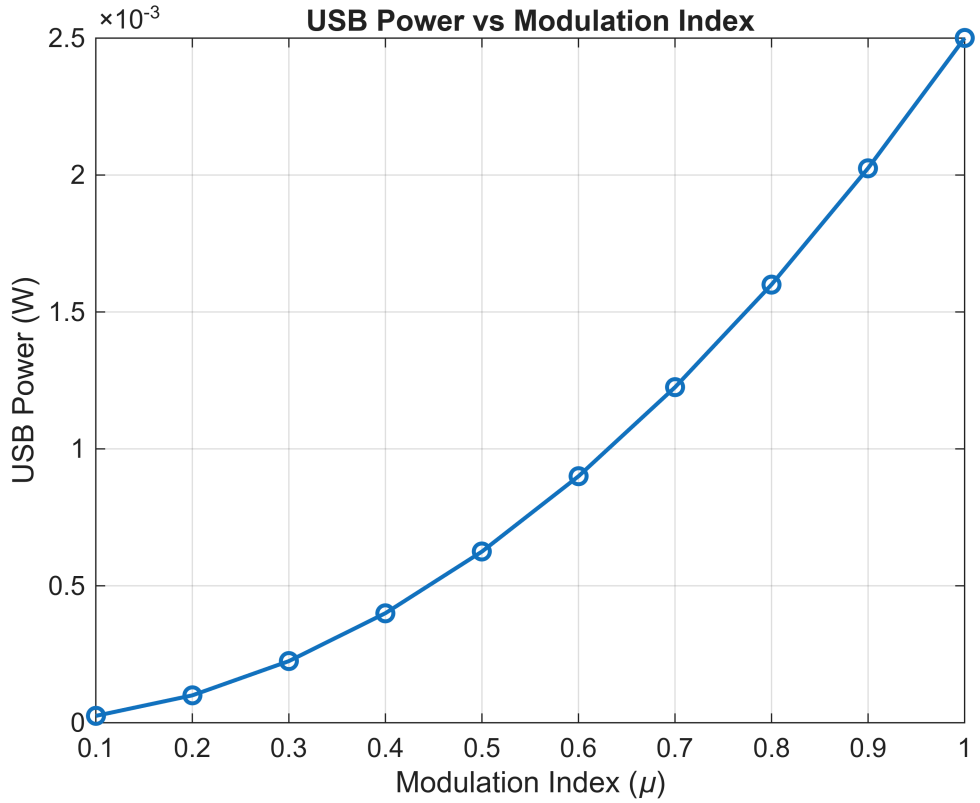
figure;

plot(mu_values,P_USB_values,'o-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('USB Power (W)');

title('USB Power vs Modulation Index');

grid on;
```




```
%% ============================================================
% 18. LSB POWER VS MODULATION INDEX
% ============================================================

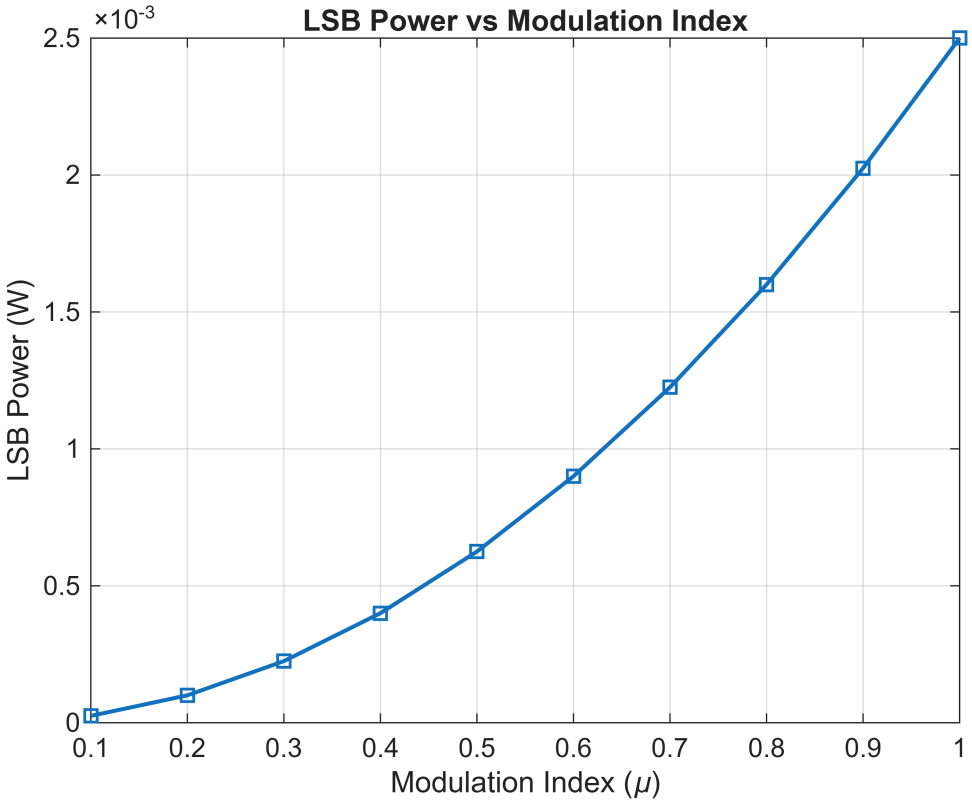
figure;

plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('LSB Power (W)');

title('LSB Power vs Modulation Index');

grid on;
```




```
%% ============================================================
% 19. TOTAL SIDEBAND POWER VS MODULATION INDEX
% ============================================================

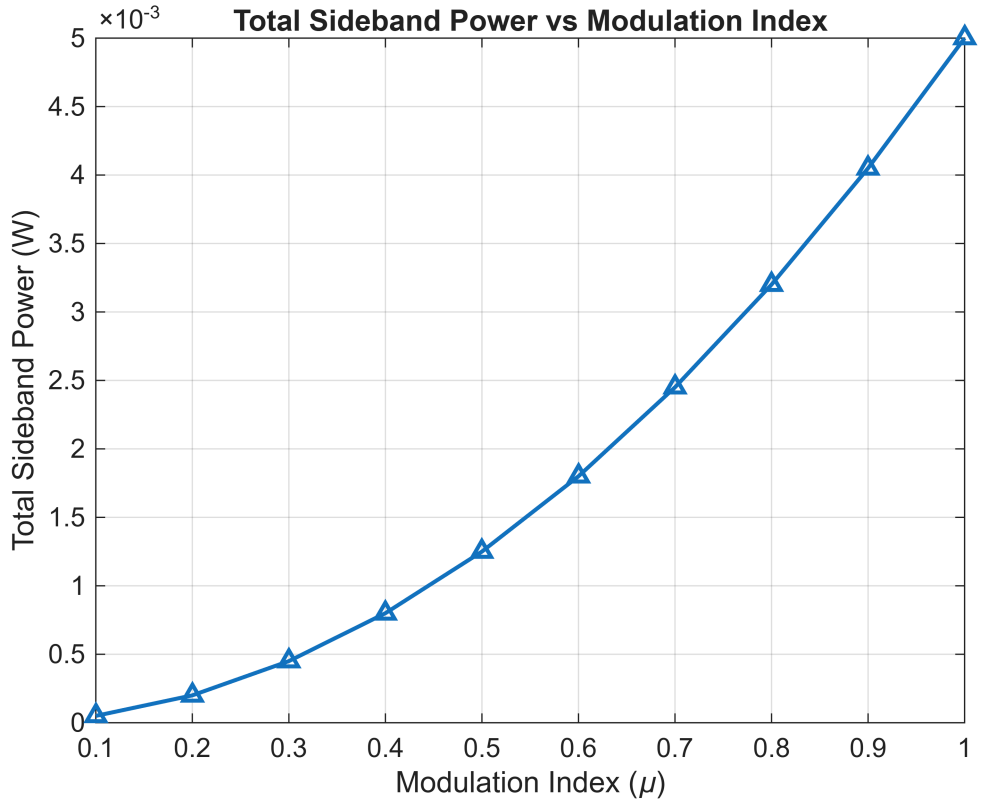
figure;

plot(mu_values,P_SB_values,'^-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Total Sideband Power (W)');

title('Total Sideband Power vs Modulation Index');

grid on;
```




```
%% ============================================================
% 20. TOTAL AM POWER VS MODULATION INDEX
% ============================================================

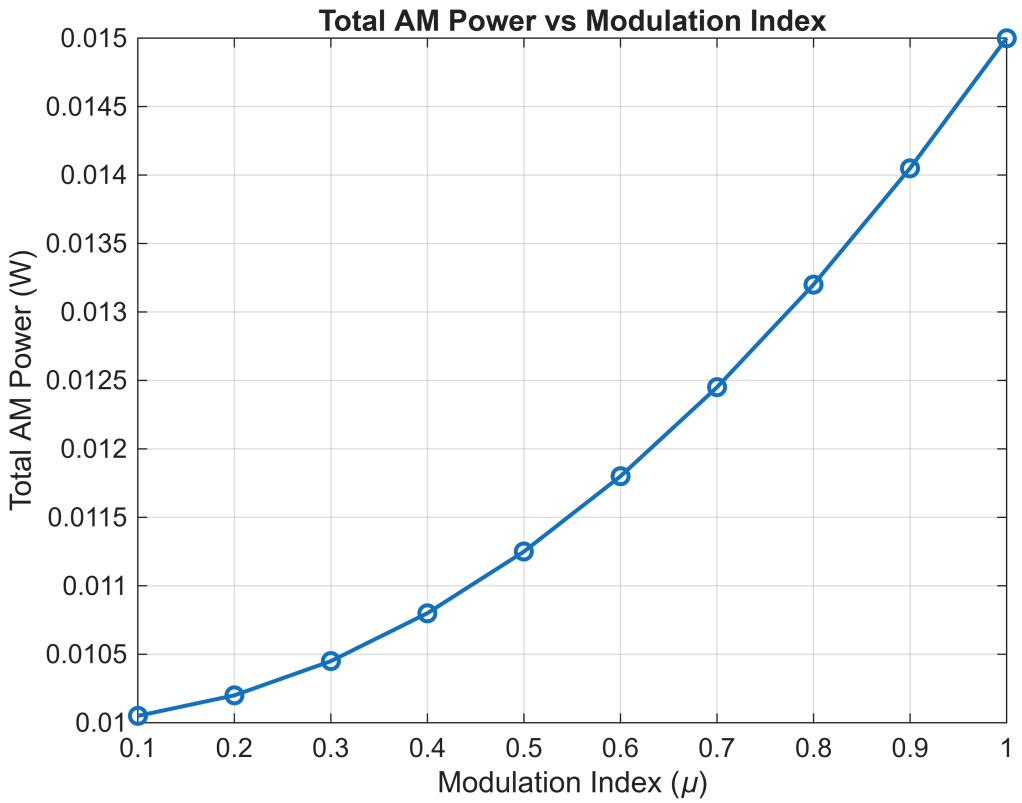
figure;

plot(mu_values,P_total_values,'o-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Total AM Power (W)');

title('Total AM Power vs Modulation Index');

grid on;
```




```
%% ============================================================
% 21. TRANSMISSION EFFICIENCY
% ============================================================

figure;

plot(mu_values,Efficiency_values,'d-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Efficiency (%)');

title('AM Transmission Efficiency vs Modulation Index');

grid on;
```




```
%% ============================================================
% 22. COMPARE USB AND LSB POWER
% ============================================================

figure;

plot(mu_values,P_USB_values,'o-','LineWidth',1.5);

hold on;

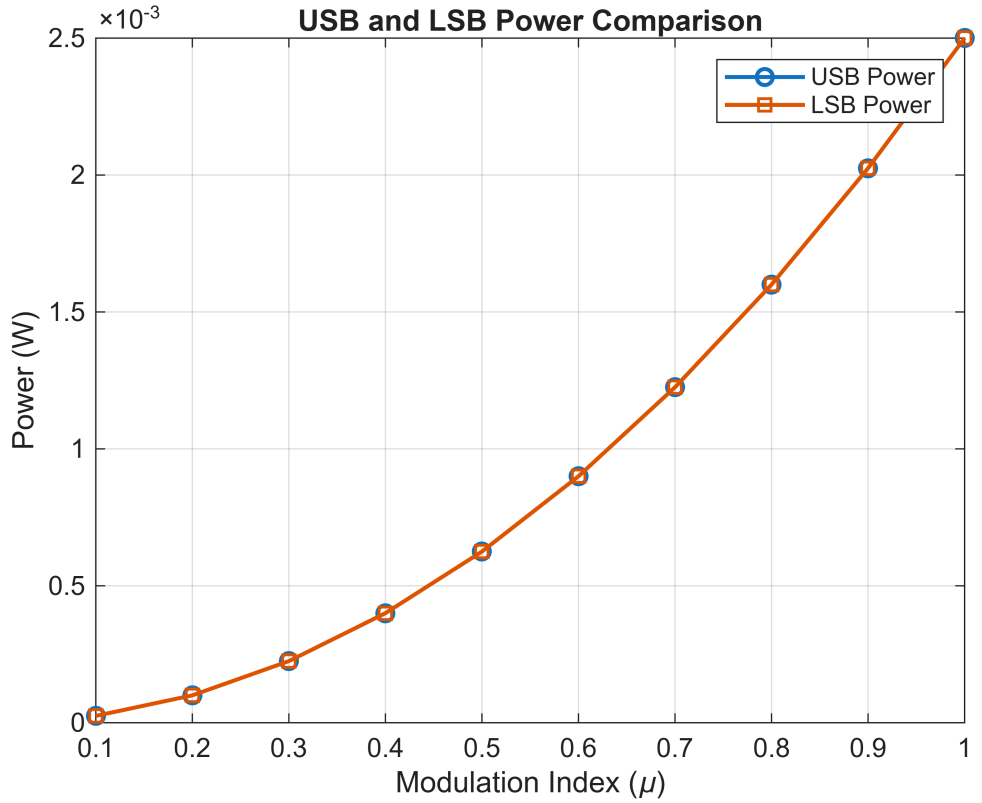
plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Power (W)');

title('USB and LSB Power Comparison');

legend('USB Power','LSB Power');

grid on;
```




```
%% ============================================================
% 23. AM SPECTRUM FOR DIFFERENT MODULATION INDICES
% ============================================================

mu_plot = [0.25 0.5 0.75 1.0];

figure;

for k = 1:length(mu_plot)

    mu_temp = mu_plot(k);

    AM_temp = Ac*(1 + ...
        mu_temp*cos(2*pi*fm*t)) ...
        .* cos(2*pi*fc*t);

    spectrum_temp = fftshift(fft(AM_temp));

    magnitude_temp = ...
        abs(spectrum_temp)/N;

    subplot(4,1,k);

    plot(f/1000,magnitude_temp);
```

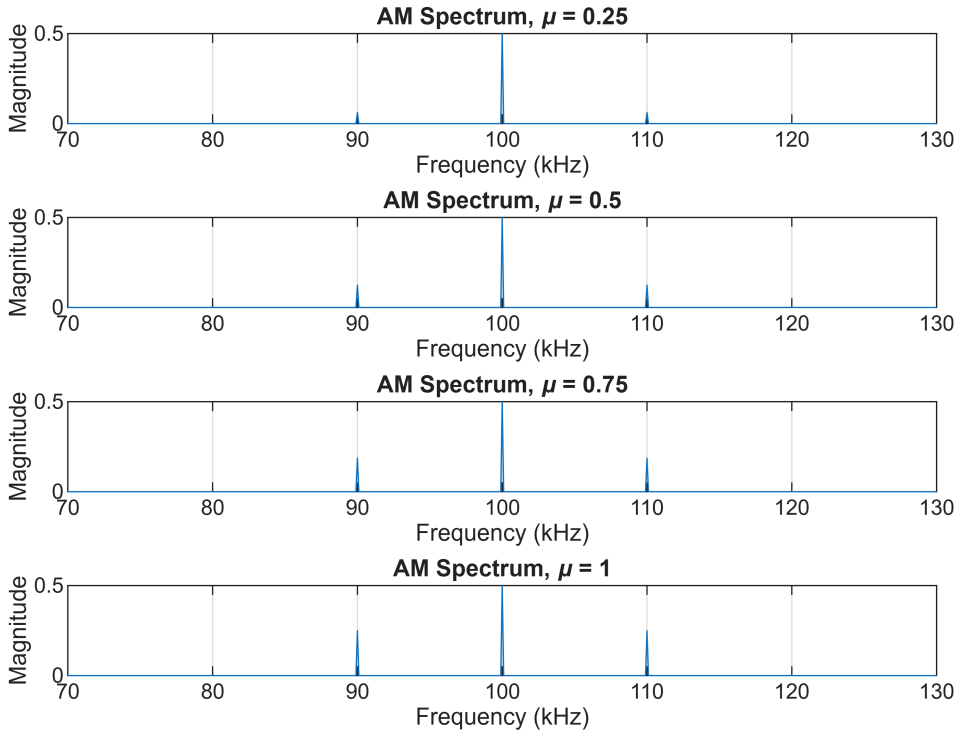
```
    xlabel('Frequency (kHz)');
    ylabel('Magnitude');

    title(['AM Spectrum, \mu = ', ...
        num2str(mu_temp)]);

    grid on;

    xlim([70 130]);

end
```




```
%% ============================================================
% 24. ENVELOPE OF AM SIGNAL
% ============================================================

envelope_upper = Ac*(1 + mu*cos(2*pi*fm*t));

envelope_lower = -Ac*(1 + mu*cos(2*pi*fm*t));

figure;

plot(t*1000,AM_signal);

hold on;
```

```
plot(t*1000,envelope_upper,'--','LineWidth',1.2);

plot(t*1000,envelope_lower,'--','LineWidth',1.2);

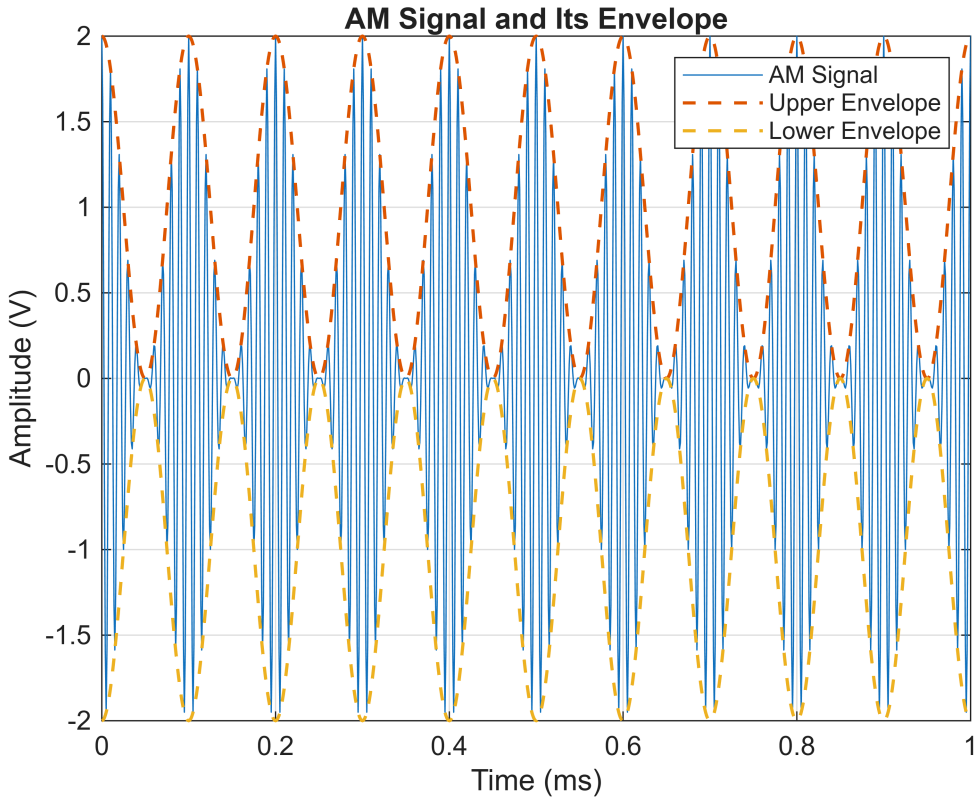
xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('AM Signal and Its Envelope');

legend('AM Signal','Upper Envelope','Lower Envelope');

grid on;

xlim([0 1]);
```




```
%% ============================================================
% 25. IMPORTANT RESULTS AT 100% MODULATION
% ============================================================

mu_100 = 1;

P_USB_100 = (mu_100^2 * Pc)/4;

P_LSB_100 = (mu_100^2 * Pc)/4;

P_SB_100 = P_USB_100 + P_LSB_100;
```

```
P_total_100 = Pc + P_SB_100;

efficiency_100 = ...
    (P_SB_100/P_total_100)*100;

fprintf('\n');
fprintf('=================================================\n');
```

```
=================================================
```

```
fprintf('          RESULTS FOR 100%% MODULATION\n');
```

```
          RESULTS FOR 100% MODULATION
```

```
fprintf('=================================================\n');
```

```
=================================================
```

```
fprintf('Carrier Power            = %.6f W\n',Pc);
```

```
Carrier Power            = 0.010000 W
```

```
fprintf('USB Power                = %.6f W\n',P_USB_100);
```

```
USB Power                = 0.002500 W
```

```
fprintf('LSB Power                = %.6f W\n',P_LSB_100);
```

```
LSB Power                = 0.002500 W
```

```
fprintf('Total Sideband Power     = %.6f W\n',P_SB_100);
```

```
Total Sideband Power     = 0.005000 W
```

```
fprintf('Total AM Power           = %.6f W\n',P_total_100);
```

```
Total AM Power           = 0.015000 W
```

```
fprintf('Transmission Efficiency  = %.2f %%\n',efficiency_100);
```

```
Transmission Efficiency  = 33.33 %
```

```
fprintf('=================================================\n');
```

```
=================================================
```

```
%% ============================================================
```

```
% 26. FINAL CONCLUSION
% ========================================================

fprintf('\n');
fprintf('CONCLUSION:\n');
```

```
CONCLUSION:
```

```
fprintf(['1. USB frequency = fc + fm.\n']);
```

```
1. USB frequency = fc + fm.
```

```
fprintf(['2. LSB frequency = fc - fm.\n']);
```

```
2. LSB frequency = fc - fm.
```

```
fprintf(['3. AM bandwidth = 2fm.\n']);
```

```
3. AM bandwidth = 2fm.
```

```
fprintf(['4. Sideband power increases with mu^2.\n']);
```

```
4. Sideband power increases with mu^2.
```

```
fprintf(['5. At 100%% modulation, total sideband power = Pc/2.\n']);
```

```
5. At 100% modulation, total sideband power = Pc/2.
```

```
fprintf(['6. At 100%% modulation, total AM power = 1.5Pc.\n']);
```

```
6. At 100% modulation, total AM power = 1.5Pc.
```

```
fprintf(['7. Maximum theoretical AM efficiency = 33.33%%.\n']);
```

```
7. Maximum theoretical AM efficiency = 33.33%.
```

```
fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('             SIMULATION COMPLETED\n');
```

```
             SIMULATION COMPLETED
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```